

GETACHEW AMBAYE, Ph.D.

ROBOTICS ENGINEER — EMBEDDED SYSTEMS — AI & AUTOMATION — MECHATRONICS

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🌐 [linkedin.com/in/getachew-ambaye](https://www.linkedin.com/in/getachew-ambaye) 🌐 Portfolio

TECHNICAL SKILLS

- **Programming:** Python, MATLAB
- **Robotics:** ABB Robots, Universal Robots (UR), RoboDK, ABB RobotStudio, Robot Programming
- **Embedded Systems:** Raspberry Pi, Microcontrollers, Sensors, IMUs, Data Acquisition
- **Controls & Automation:** Siemens PLC, CLICK PLC, Electro-Pneumatics, Hydraulics
- **AI & Computer Vision:** Machine Learning, Deep Learning, OpenCV, Image Processing
- **CAD/Simulation:** SolidWorks, CATIA V5, Autodesk Inventor, AutoCAD, ABAQUS, ANSYS, MSC ADAMS
- **Manufacturing:** Additive Manufacturing (3D printer), Rapid Prototyping, GD&T, Product Development

PROFESSIONAL EXPERIENCE

- Assistant Professor, Engineering Technology** 2025–Present
Austin Peay State University
- Robotics, machine vision, embedded systems, and automation projects.
 - PLC programming, sensor integration, rapid prototyping, and VR applications.
- Graduate Research Assistant** 2022–2025
Wichita State University
- Machine learning, robotics, soft robotics, computer vision, and predictive diagnostics.
 - Finite element simulation, experimental validation, and intelligent manufacturing research.
- Engineering Faculty** 2024–2025
Hesston College
- Mechatronics, automation, product development, testing, and validation.

SELECTED ENGINEERING PROJECTS

- Soft Robot Workspace Prediction Using AI**
- Developed machine-learning models to predict soft robotic actuator workspaces using FEA data.
- Robot Arm Damage Detection**
- Developed deep-learning models using vibration signals for robotic health monitoring.
- Machine Vision Inspection System**
- Developed computer vision algorithms for automated drilled-hole misalignment detection.
- Embedded Diagnostic Platform**
- Integrated RPM, vibration, displacement, and IMU sensing into a real-time diagnostic system.
- VR Engineering Training Platform**
- Developed immersive VR environments for engineering assembly and visualization.

EDUCATION

- Ph.D., Industrial Engineering** Wichita State University, 2025
M.Sc., Mechanical Design Engineering Jimma University, 2020
B.Sc., Mechanical Engineering Jimma University, 2016

SELECTED PUBLICATIONS

- Soft Robot Workspace Estimation via Finite Element Analysis and Machine Learning, Actuators, 2025.
- Robot Arm Damage Detection Using Vibration Data and Deep Learning, Neural Computing and Applications, 2024.
- Soft Robot Design, Manufacturing, and Operation Challenges: A Review, JMMP, 2024.

CERTIFICATIONS

- ASME GD&T Drawing Interpretation
- ASME Nondestructive Examination (Section V)
- IISE Lean Green Belt
- Higher Diploma Program in Engineering Education

AWARDS & MEMBERSHIPS

- Outstanding Graduate Student Award (2025)
Best Paper Award, EAI ICAST (2023)
IEEE — ASME — ASEE — ASNT — Alpha Pi Mu